



A comprehensive survey on 6G: Enabling technologies, key applications, and future challenges

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ARTICLE INFO

Keywords:

5G/6G
Emerging technologies
Wireless communication
Mobile communication
Survey

ABSTRACT

The rapid evolution of wireless communication technologies from 1G to 5G has enabled remarkable advances in connectivity, yet emerging applications such as holographic telepresence, autonomous systems, wireless brain-computer interfaces, and ubiquitous 3D coverage expose fundamental limitations in current networks. While numerous surveys have explored individual 6G enablers, few studies provide a unified and comprehensive analysis that integrates the technological foundations, architectural transformations, and multidisciplinary applications required to realize future 6G systems. This manuscript addresses this gap by offering a holistic overview of 6G, beginning with the motivation for moving beyond 5G and the performance demands driving next-generation research. It synthesizes key enabling technologies including terahertz communication, ultra-massive MIMO, intelligent reflective surfaces, AI-native networking, Digital Twin Networks, and generative artificial intelligence. The paper further examines new architectural models spanning terrestrial, aerial, satellite, and underwater networks, while also discussing emerging 6G use cases across healthcare, robotics, industry, mobility, and immersive environments. Finally, the manuscript outlines open challenges and research directions that must be addressed to transform the 6G vision into practical reality. By bridging foundational concepts and forward-looking innovations, this survey provides a comprehensive reference for researchers and practitioners seeking to understand and contribute to the evolving 6G landscape.

1. Introduction

The evolution of mobile communication networks has been one of the most transformative technological journeys of the modern era [1]. Since the deployment of the first generation (1G) cellular networks in the 1980s, each subsequent generation, 2G, 3G, 4G, and more recently 5G, has introduced significant advancements in data rate, latency, network capacity, and service capabilities. These generational shifts have not only enhanced user experience but also catalyzed the proliferation of mobile-based services and digital ecosystems. Driven by rapid advancements in hardware, spectrum utilization, and protocol design, the mobile network landscape has continuously evolved to meet the ever-growing demand for connectivity and mobility in both personal and industrial domains [2].

Fifth-generation (5G) networks represent a considerable leap forward in wireless communication, introducing novel features such as enhanced Mobile Broadband (eMBB), Ultra-Reliable and Low-Latency Communications (URLLC), and Massive Machine-Type Communications (mMTC) [3]. These features have supported a wide range of applications including augmented and virtual reality, smart city infrastructures, remote healthcare, and Industry 4.0 paradigms. However,

despite these capabilities, 5G networks are increasingly facing limitations in addressing the demands of next-generation use cases that require ultra-high reliability, extremely low latency, unprecedented data rates, and intelligent, context-aware networking capabilities [4].

The projected demands of the 2030s, such as holographic telepresence, tactile internet, wireless brain-computer interfaces, and the integration of artificial intelligence (AI) into every layer of the network, have underscored the necessity for a new paradigm in mobile communications [5]. The sixth generation (6G) is envisioned to transcend the foundational capabilities of 5G by incorporating Terahertz (THz) communications, intelligent reflective surfaces, quantum and blockchain technologies, AI-native network management, and non-terrestrial networks (NTNs), thereby enabling global, three-dimensional connectivity with seamless interoperability [6]. The development of 6G aims not only to address the shortcomings of current systems but also to anticipate future societal, industrial, and technological demands in a sustainable and inclusive manner [7].

The contributions of this work can be summarized across several key dimensions. First, the manuscript provides a comprehensive motivation for the shift beyond 5G, clearly outlining the current limitations of

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Table 1

Comparison of representative 6G surveys and their relation to this work.

Survey	Year	Scope/Focus	Limitations compared to this work
Huang et al. [8]	2019	Green 6G networks, energy-efficient design, early architecture concepts.	Focuses mainly on sustainability; does not cover modern enablers such as DTN, generative AI, or concrete use-case workflows.
Uusitalo et al. (Hexa-X) [9]	2021	European 6G vision, key use cases, high-level enabling technologies.	Visionary but conceptual; lacks detailed enabling-technology comparisons and practical 6G application demonstrations.
Bassoli et al. [10]	2021	Motivation for 6G and limitations of 5G; early research drivers.	Strong on motivation but limited coverage of technologies, architectures, and integrated 6G system design.
Salameh & El Tarhuni [11]	2022	Overview of 5G→6G evolution, challenges, and emerging technologies.	Limited architectural depth and does not integrate AI-native networks, DTN, or practical mission-critical workflows.
Hakeem et al. [12]	2022	Vision and research directions; identifies challenges and enablers.	Does not provide a unified comparison of technologies, architectures, and detailed real-world scenarios.
Shafi et al. [2]	2024	High-level evolution towards 6G; key performance indicators and trends.	Technology review remains broad; does not explore integrated architectures or detailed use-case implementations.
This work	2025	Comprehensive survey covering 6G technologies, architectures, applications, DTN, GenAI, and practical workflows.	Provides unified analysis of enabling technologies, detailed 3D architectures, realistic holographic-surgery scenario, and clear research directions, offering greater breadth and depth than prior surveys.

existing networks and the emerging demands that justify the need for 6G. Building on this foundation, the survey offers a holistic and structured review of essential 6G enabling technologies, including terahertz communication, ultra-massive MIMO, intelligent reflective surfaces, AI-native networking, edge intelligence, Digital Twin Networks, and generative artificial intelligence, emphasizing their roles, advantages, and open research challenges. In addition, the paper delivers a detailed analysis of next-generation network architectures by presenting an integrated perspective of space–air–ground–sea systems, highlighting architectural design principles, major components, interoperability constraints, and system-level considerations. The manuscript further examines a broad set of emerging 6G application domains such as extended reality, autonomous systems, intelligent healthcare, the Internet of Everything, and holographic communication. To illustrate the practical relevance of these technologies, a realistic end-to-end scenario of 6G-enabled remote holographic surgery is presented, demonstrating how multiple 6G enablers converge to support mission-critical operations. Finally, the survey synthesizes the major technical challenges and research opportunities associated with the development of 6G networks, identifying open problems and outlining promising directions for future investigation. Taken together, these contributions position this work as a comprehensive and forward-looking reference for both academic researchers and industry practitioners engaged in shaping the next generation of wireless communication systems.

This paper presents a comprehensive survey of 6G mobile networks, focusing on the technological trajectory that has led to its emergence, and the fundamental requirements it seeks to fulfill. The paper is structured as follows: Section 2 provides an overview of the historical evolution of mobile networks from 1G to 5G, highlighting the key advancements and limitations that have motivated the shift towards 6G. Section 3 outlines the technological motivations and conceptual vision of 6G, including performance objectives and expected capabilities. Section 4 introduces and analyzes the enabling technologies that will form the backbone of 6G infrastructure, such as Terahertz communication, edge intelligence, massive MIMO, and quantum-enhanced security frameworks. Section 5 discusses the anticipated changes in core network architecture, emphasizing the need for a flexible, scalable, and intelligent design. Section 6 explores the diverse use cases and application scenarios that 6G aims to support, ranging from immersive extended reality environments to autonomous systems and intelligent healthcare. Section 7 outlines the major technical and theoretical challenges that must be addressed to realize the full potential of 6G networks. Finally, the concluding section summarizes the key insights and offers directions for future research. This structure is illustrated in Fig. 1.

By synthesizing current research trends and future projections, this paper aims to provide a holistic understanding of the forthcoming 6G era and to contribute to the ongoing academic and industrial discourse on next-generation mobile communications.

Although several survey papers have examined different aspects of 6G, their coverage varies significantly in terms of scope, depth, and technological focus. Table 1 summarizes these works and clearly demonstrates how this manuscript provides a broader and more integrated perspective, covering enabling technologies, architectures, applications, and practical use-case workflows.

2. Mobile network evolution

Mobile networks have undergone a remarkable and rapid evolution since their inception in the 1980s. This continuous evolution has given rise to several generations of mobile telephony, with each succeeding generation being introduced approximately every decade [13].

Fig. 2 provides a visual representation of the progression of mobile network evolution.

The Table 2 presents a comparative overview of mobile network generations, highlighting their main features, technologies and applications.

2.1. First steps from 1G to 3G

The inception of cellular networks dates back to the 1980s, when the first generation was introduced, primarily designed for voice services and capable of data rates up to 2.4 kbps. In contrast to the analog systems of the first generation, the second generation embraced digital modulation technologies like Time Division Multiple Access (TDMA). This technological shift brought about a significant enhancement, offering data rates of up to 64 kbps and introducing features such as short message service (SMS) [14]. The dawn of the third generation in the year 2000 marked a central moment in mobile network evolution, aiming to provide high-speed data transmission. The 3G network was a game-changer, offering a minimum data transfer rate of 2 Mbps and granting users access to the vast realm of the internet. This transformation enabled an array of advanced services previously unsupported by the 1G and 2G networks, including web browsing, TV streaming, navigation maps, and video services [15].

2.2. 4G evolution

The advent of 4G marked a significant milestone in the evolution of mobile networks, introduced in the late 2010s [13]. This all-IP network

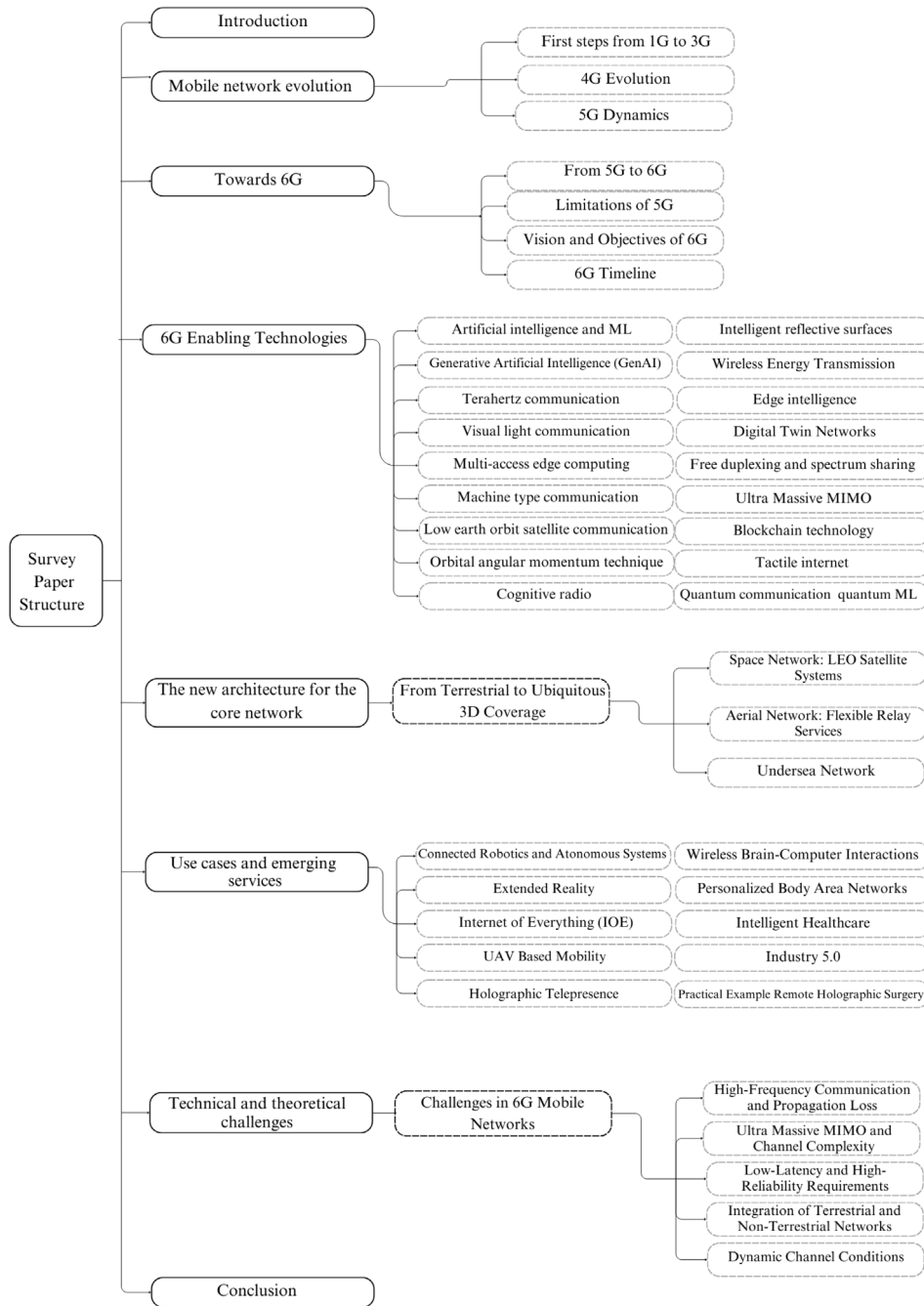


Fig. 1. Structure of the paper with main sections and flow.

was designed to deliver exceptionally high-speed data rates, reaching up to 1 Gbps for downlink and 500 Mbps for uplink transmissions. Its core enhancements included improvements in spectral efficiency and a reduction in latency, making it well-suited to meet the requirements of demanding applications such as digital video broadcasting (DVB), high-definition TV content delivery, and seamless video chat experiences [6]. Notably, Long Term Evolution-Advanced (LTE-A) and Wireless Interoperability for Microwave Access (WiMax) were both recognized as 4G standards. The foundation of 4G, LTE, integrated a blend of existing and novel technologies, including Coordinated Multi-Point (CoMP), multiple input multiple output (MIMO), and orthogonal frequency division multiplexing (OFDM) [16].

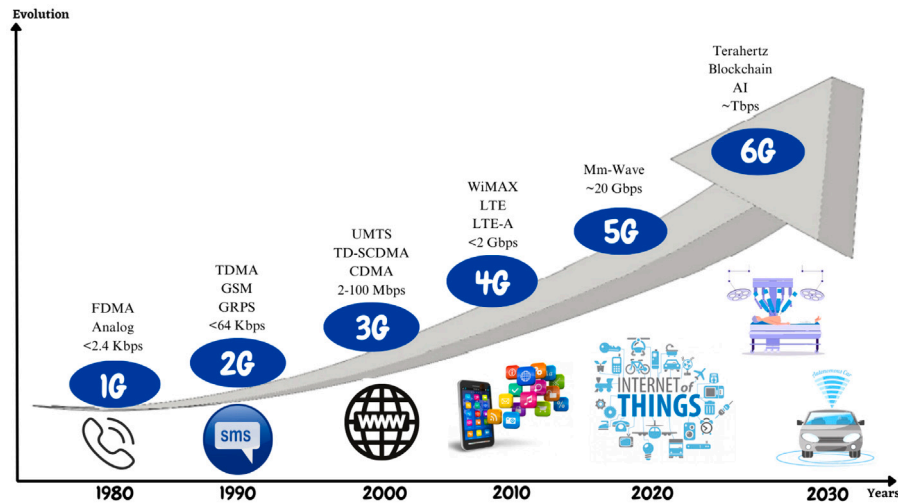
2.3. 5G dynamics

The primary objective of 5G technology is to achieve revolutionary advances in terms of throughput, latency, network reliability, and energy efficiency, all while ensuring extensive connectivity. Unlike its predecessors, 5G harnesses not only the new sub-6 GHz spectrum, which includes the C-band and the broader 6 GHz band but also, for the first time, delves into the millimeter-wave band, enabling significantly higher data rates, reaching up to an impressive ten Gbps [18]. 5G relies on advanced access technologies, including beam division multiple access (BDMA), real-time division multiple access, and filter bank multiple access (FBMC). To enhance network performance, 5G incorporates

Table 2

Comparison of mobile network generations from 1G to 6G.

Feature	1G	2G	3G	4G	5G	6G
Introduction	1980s	1990s	2000s	2010s	2020s	2030s (Expected)
Technology	Analog	Digital	Packet-Switched	LTE (IP-based)	NR (New Radio)	AI-driven, THz frequencies
Data rate	2.4 kbps	64 kbps–200 kbps	2 Mbps–42 Mbps	100 Mbps–1 Gbps	10 Gbps	1 Tbps (Expected)
Latency	High (600 ms)	300–600 ms	100–500 ms	10–50 ms	1–10 ms	<1 ms
Multiple access	FDMA	TDMA, CDMA	CDMA, WCDMA	OFDMA, SC-FDMA	OFDMA, mmWave	AI-Driven, Reconfigurable Intelligent Surfaces (RIS)
Key features	Voice calls	SMS, Basic data	Mobile internet, Video calls	High-speed internet, VoLTE	Ultra-reliable, Low latency, Massive IoT	AI-Integrated, Quantum communication
Frequency bands	800 MHz	900 MHz, 1800 MHz	850 MHz–2100 MHz	700 MHz–2.5 GHz	Sub-6 GHz, mmWave (24–100 GHz)	THz band (>100 GHz)
Main applications	Voice communication	SMS, Basic internet	Mobile web, Video streaming	HD streaming, IoT, VoLTE	Smart cities, IoT, Industry 4.0	Holographic communication, Tactile internet, Smart implants
Challenges	Poor security, Low capacity	Limited data rate	Spectrum congestion	Battery drain, Network congestion	Infrastructure cost, Deployment complexity	Energy efficiency, THz signal penetration

**Fig. 2.** Evolution of mobile wireless systems [11].

a multitude of emerging technologies. These innovations encompass massive MIMO to boost network capacity, software-defined networks (SDN) for increased network flexibility, device-to-device (D2D) communication to improve spectral efficiency, information-centric networking (ICN) to reduce network traffic, and network slicing to facilitate the swift deployment of numerous services [17].

Within the framework of IMT 2020, three primary usage scenarios have been proposed for 5G technology: enhanced mobile broadband (eMBB) to cater to high-speed data requirements, ultra-reliable low-latency communications (URLLC) for mission-critical applications, and massive Machine Type Communications (mMTC) to support the proliferation of connected devices and the Internet of Things like presented in Fig. 3.

3. Towards 6G

3.1. From 5G to 6G: Why do we need a new generation?

The evolution from 5G to 6G is driven by the ever-increasing demands for higher speeds, lower latencies, and greater connectivity

density. While 5G has brought unprecedented capabilities, such as enhanced mobile broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive machine-type communications (mMTC), the growing complexity of applications in industries, health, education, and entertainment has revealed the need for an even more advanced network [19]. Applications such as holographic telepresence, and extended reality (XR) are expected to require terabit-per-second (Tbps) data rates and latencies below 0.1 ms demands beyond 5G's capabilities [20]. The 6G era, expected to emerge around 2030, seeks to address these requirements by not only increasing network performance but also enabling new paradigms of connectivity that integrate advanced technologies such as artificial intelligence (AI), machine learning (ML), and quantum communication into the core of the network [21].

3.2. Limitations of 5G in addressing future needs

Although 5G represents a major advancement over previous mobile generations, it still falls short of meeting the requirements of future intelligent communication systems. One fundamental limitation lies in

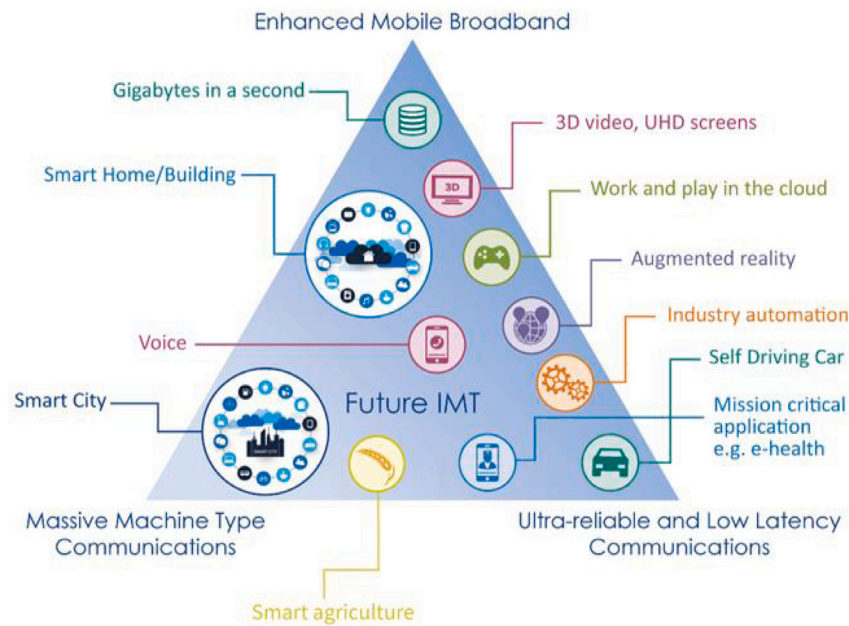


Fig. 3. The different 5G usage [17].

spectrum availability: even with the use of sub-6 GHz and millimeter-wave bands, 5G cannot provide the ultra-wide bandwidth necessary for emerging applications such as holographic telepresence, multi-user XR, and terabit-scale data services, which demand orders of magnitude more capacity than current deployments can offer [16,22]. Additionally, the dense small-cell infrastructure required for high-frequency operation significantly increases network energy consumption, raising sustainability concerns as device numbers continue to grow [23]. Another major constraint is 5G's inability to support ultra-massive connectivity at the scale expected for 6G, where billions of heterogeneous devices, including sensors, machines, autonomous robots, and intelligent surfaces, must coexist with strict reliability guarantees [24]. Latency is also a major bottleneck: despite achieving sub-millisecond performance under ideal conditions, 5G cannot consistently deliver the deterministic, end-to-end ultra-low latency required for real-time applications such as remote surgery, tactile Internet, or holographic communication [17]. Moreover, 5G lacks native support for AI-driven networking, limiting its ability to autonomously manage traffic flows, predict network behavior, or optimize resource allocation in complex, dynamic environments [25]. These limitations collectively illustrate the need for a new generation of wireless systems 6G, that can address spectrum scarcity, energy efficiency, connectivity density, intelligent operation, and extreme latency requirements.

3.3. Vision and objectives of 6G

The vision for 6G centers on creating an intelligent, ultra-reliable, and fully pervasive communication ecosystem capable of supporting the increasingly demanding applications of the next decade. Unlike 5G, which primarily enhances speed and connectivity, 6G aims to deliver terabit-per-second data rates, sub-0.1 ms latency, and globally integrated coverage by leveraging new spectrum ranges such as the sub-THz and THz bands [26]. A key ambition of 6G is to enable seamless end-to-end communication across terrestrial, aerial, satellite, and underwater systems, thereby ensuring ubiquitous connectivity even in remote or previously inaccessible regions [27]. Beyond raw performance improvements, 6G introduces a paradigm shift towards AI-driven network automation, where machine learning and distributed intelligence are embedded at every layer of the network to optimize resource management, predict traffic patterns, enhance reliability, and

enable autonomous operation. These capabilities are essential for supporting advanced and highly interactive applications such as extended reality (XR), autonomous and collaborative robotic systems, digital twins, and holographic communication, all of which require extreme data rates, deterministic latency, and real-time context awareness [28]. By integrating communication, sensing, computing, and AI into a unified framework, 6G aspires to create a transformative platform that not only extends human-to-human and human-to-machine interaction but also enables emerging intelligent ecosystems at global scale.

3.4. 6G timeline

Each generation of wireless communications technology typically spans around ten years before moving on to the next. 5G's journey began with its initial standard published in 2018 as part of 3GPP Release 15, marking the foundations of the current generation. Since then, 5G has evolved progressively, with subsequent releases introducing advanced capabilities. 3GPP Release 18, published in 2024, represent the next phase of 5G evolution, known as 5G Advanced. This phase aims to refine existing 5G functionalities while integrating new features to meet emerging demands [29].

Simultaneously, research and development efforts for the next-generation 6G systems are intensifying. The International Telecommunication Union (ITU), is expected to publish the IMT-2030 framework around 2026. This framework will outline the vision, objectives, and technical requirements for 6G, serving as a blueprint for future development.

Following the release of the IMT-2030 document, the 3rd Generation Partnership Project (3GPP), responsible for defining detailed specifications for mobile networks, will begin the development of the 6G standards. These specifications, designed to align with the ITU's vision, are projected to emerge between 2028 and 2030. This timeline is presented in Fig. 4 It aligns with the anticipated commercial deployment of 6G systems around 2030, ushering in a new era of wireless communication with unprecedented capabilities [30].

4. 6G enabling technologies

The realization of 6G communication networks will depend on a range of cutting-edge technologies designed to overcome the limitations of previous generations and meet the unprecedented demands



Fig. 4. Estimated timeline of standardization and delivery for 6G networks [31].

of the future. These enabling technologies are pivotal in achieving the extreme data rates, ultra-low latencies, and ultra massive connectivity envisioned for 6G systems. Below, we explore these technologies and their contributions.

4.1. Artificial intelligence and machine learning

AI is set to become one of the foundations of 6G networks, enabling intelligent decision-making and optimization across the network. From resource allocation to interference management, AI will empower 6G systems with self-learning and adaptive capabilities. Key applications include autonomous network configuration, anomaly detection, and predictive maintenance. AI facilitates enhanced user experience through personalized services and dynamic quality-of-service (QoS) adjustments. AI also enables closed-loop optimization to wireless 6G networks. In this technique, the user acts in accordance with device feedback. By maintaining end-to-end collective optimization at the system layer, AI can further be implemented to optimize wireless communication systems. This intelligent layer offers numerous possibilities, such as self-optimization, self-learning, and data collection. These examples are only a small part of the widespread applications of AI technologies and the innovations they will bring to the 6G network structure [32].

Machine learning, a subset of AI, will play an important role in analyzing and processing the massive amounts of data generated by 6G networks. ML algorithms will optimize traffic management, energy efficiency, and user mobility prediction. Reinforcement learning models will be used for real-time decision-making in scenarios such as spectrum allocation and interference mitigation [33].

4.2. Generative Artificial Intelligence (GenAI)

Generative Artificial Intelligence (GenAI) is poised to play a critical role in 6G, enabling the creation, prediction, and synthesis of complex network information using advanced generative models such as GANs, VAEs, diffusion models, and large language models (LLMs). Unlike traditional AI, which primarily performs classification or regression tasks, GenAI can generate realistic channel models, mobility patterns, network traffic predictions, synthetic training datasets, and even optimized communication waveforms [34].

In 6G physical-layer design, generative models enable the reconstruction of high-dimensional radio environments, allowing more accurate channel estimation and improved beamforming for THz and mmWave systems. GenAI also accelerates the creation of realistic digital twins by producing synthetic but reliable network states, thereby reducing data collection overhead [35].

At the network layer, GenAI facilitates proactive resource allocation, anomaly detection, and autonomous decision-making. Furthermore, generative models are essential for security in 6G, enabling adversarial training, synthetic attack simulation, and privacy-preserving data generation.

Recent publications demonstrate the potential of GenAI in communication systems, including intelligent waveform design, generative channel modeling, and zero-shot learning for network optimization. As 6G evolves towards AI-native architectures, GenAI will be fundamental for enabling self-evolving, self-healing, and self-optimizing networks [36].

4.3. Terahertz communication

The Terahertz (THz) band, presented in Fig. 5, spanning 0.1–10 THz, represents the last untapped region in the radio frequency spectrum, offering ultra-high bandwidth and data rates up to 1 Tbps. It bridges the gap between millimeter-wave (mmWave) and infrared (IR) frequencies, combining broader transmission bandwidth than mmWave with better propagation characteristics than IR. THz communication is essential for 6G, enabling applications such as holographic telepresence, and ultra-HD streaming [37].

With its ability to support ultra-dense networks and massive capacity, THz communication is poised to meet 6G's demands for unprecedented data rates and low latency. Advantages include higher spectral efficiency, integration of thousands of small antenna elements into base stations, and enhanced security through high directional beams. However, challenges like system complexity, power consumption, and THz component design must be addressed. Leveraging advancements like ultra-massive MIMO and Intelligent Reflective Surfaces, THz communication is a promising enabler for 6G networks [38].

4.4. Visual light communication

Visual Light Communication (VLC) uses visible light as a wireless communication medium, offering high data rates, immunity to electromagnetic interference, and secure, interference-free communication. It is especially suitable for indoor environments like hospitals and factories, and it complements traditional systems [39].

VLC operates similarly to fiber optics, delivering fiber-like transmission rates and bandwidth through light. With terahertz-level bandwidth, high-frequency reuse, and unlicensed spectrum, VLC supports gigabit-level transmission. However, current systems face challenges like narrow LED bandwidths and the need for efficient ultra massive MIMO designs.

Despite these hurdles, advancements in VLC are expected to enable terabit-level data rates for 6G by 2030, utilizing thousands of LED active sources for short-range, high-capacity communication [40].

The Fig. 6 shows an example of inside visible light communication.

4.5. Multi-access edge computing (MEC)

Multi-access Edge Computing (MEC) enhances network efficiency by bringing computational resources closer to end users, enabling low-latency, real-time processing for applications like autonomous vehicles and augmented reality (AR). By deploying computing, caching, and data analysis at the network edge, MEC significantly reduces end-to-end latency and optimizes resource management.

In 6G, MEC will integrate application-oriented capabilities into the network, supporting AI-driven services and creating dynamic, self-optimizing machine-type communication (MTC) networks (see next in Section 4.6). With features like intelligent data elimination and network slicing, MEC ensures flexibility, localized data processing, and energy-efficient operations. These advantages position MEC as an essential technology for achieving the performance and scalability demands of 6G networks [41].

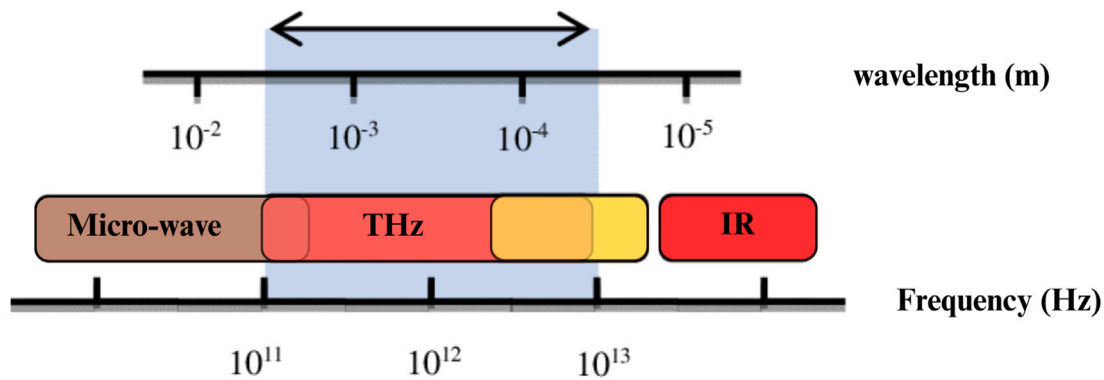


Fig. 5. The Terahertz (THz) band [38].

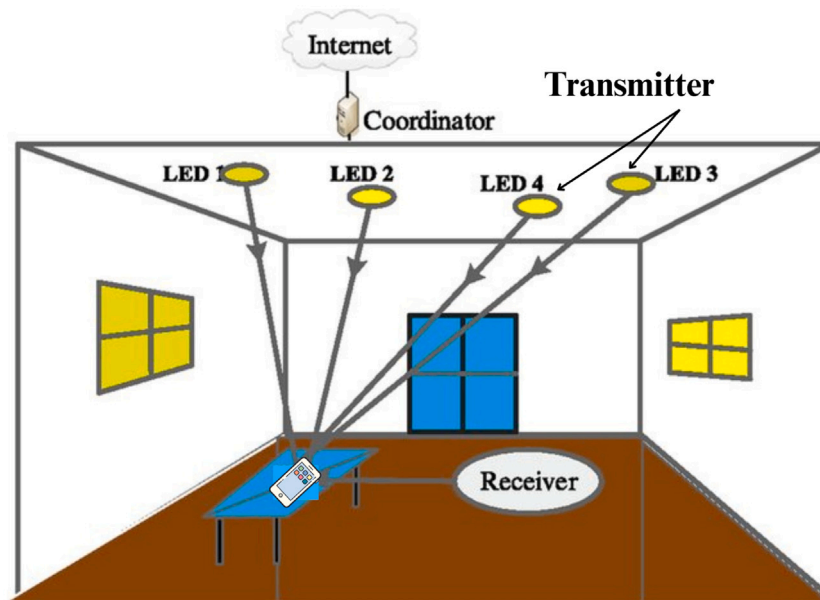


Fig. 6. Indoor visible light communication [40].

4.6. Machine type communication

6G is envisioned as the first wireless standard to offer seamless integration across diverse industry sectors, addressing the growing demand for reliable and scalable communication. It will facilitate digital transformation through vertical applications enabled by machine-type communication (MTC) [42]. MTC, which allows autonomous device-to-device communication, is crucial for IoT but poses challenges related to latency and reliability. Technologies such as Network Function Virtualization (NFV) and Software-Defined Networking (SDN) are used to address these challenges by providing flexible, efficient, and secure network infrastructures. A dynamic and unified core network will be essential for supporting the diverse requirements of 6G verticals. Furthermore, massive MTC (mMTC) will enable efficient low-power communication among billions of devices, reinforcing 6G's role as a foundational element for future intelligent networked systems [43].

4.7. Low earth orbit satellite communication

The integration of satellite capabilities with terrestrial networks in 6G is expected to address essential challenges such as efficient data transmission, resource optimization, disaster recovery, and global connectivity. Low Earth Orbit (LEO) satellites are key enablers of this vision, offering high-speed, low-latency communication and extended

coverage [44]. By supporting frequency reuse and enabling three-dimensional coverage in conjunction with terrestrial and aerial platforms, LEO networks enhance network scalability. Although satellite communication faces obstacles like delay and path loss, the adoption of mmWave technology provides high bandwidth and directivity for effective point-to-point links.

Emerging satellite architectures and heterogeneous networks further reduce deployment costs and improve coverage. Enabling technologies such as network slicing, NFV, and SDN support the seamless convergence of terrestrial and non-terrestrial infrastructures, enhancing automation, flexibility, and resource allocation. Edge computing reinforces this integration by improving mobility support, Quality of Service (QoS), and communication resilience. With all this advantage, LEO satellites play a foundational role in realizing the 6G vision of ubiquitous, reliable, and inclusive global connectivity [45].

4.8. Orbital angular momentum technique

Orbital Angular Momentum (OAM) is an emerging modulation technique that leverages the angular momentum of electromagnetic waves to transmit multiple data streams over the same frequency, thereby enhancing spectral efficiency and throughput. By utilizing helical phase wavefronts, OAM enables multiple access and robust frequency sharing,

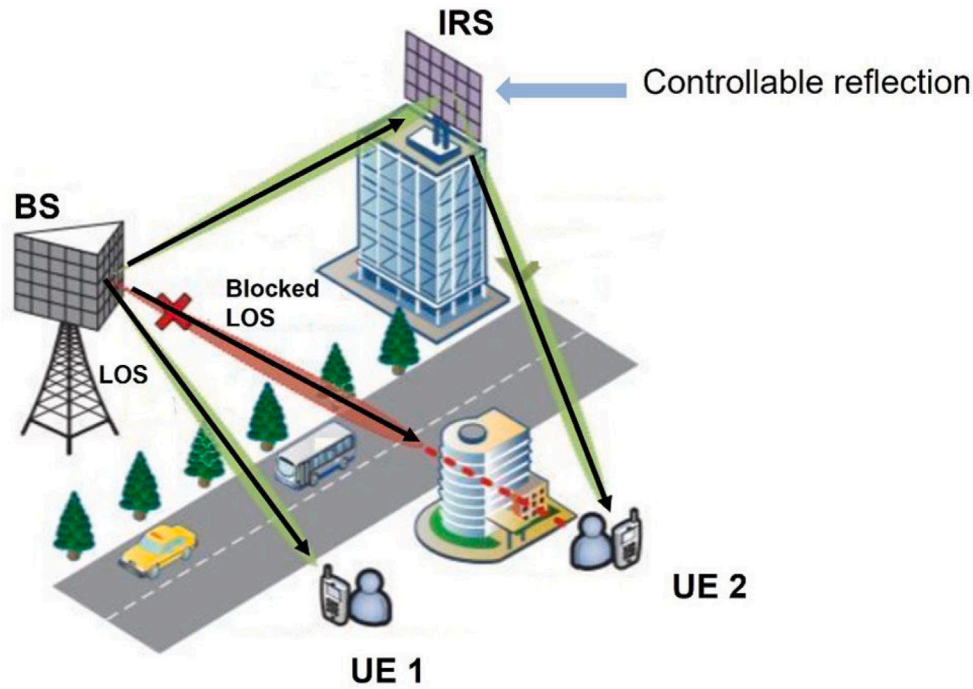


Fig. 7. Reconfigurable intelligent surfaces help ensure user equipment gets the highest received signal power [46].

making it a promising candidate for 6G networks [47]. Key advantages include high spectral efficiency, anti-jamming capabilities, and improved bandwidth utilization.

Despite its potential, OAM faces challenges such as beam divergence, which limits long-distance transmission, and sensitivity to misalignment, especially in non-line-of-sight (NLOS) conditions. Initially developed for optical systems, OAM has since been extended to Terahertz, mmWave, and lower frequency domains. Integration with advanced channel coding and frequency-sharing techniques is anticipated to enhance its viability. As research progresses, OAM is expected to be essential in overcoming bandwidth constraints and boosting 6G network performance [47].

4.9. Cognitive radio

Cognitive Radio (CR) is a dynamic spectrum access technology that enhances spectral efficiency by detecting and utilizing under-used frequency bands, thereby minimizing interference. In 6G, CR is expected to be instrumental in managing ultra-dense network deployments and maintaining seamless connectivity in complex and dynamic environments. Building on CR, Symbiotic Radio (SR) introduces a cooperative paradigm in which multiple subsystems within a heterogeneous network intelligently interact to optimize resource utilization and transmission efficiency [48].

While SR holds promise for improving overall network performance, it remains in its developmental stages, with challenges such as energy efficiency and effective spectrum optimization. The integration of CR, SR, and complementary technologies, such as inter-device communication, spectrum sharing, and full-duplex operation, will be important for enabling flexible and intelligent spectrum management in 6G systems.

4.10. Intelligent reflective surfaces

Intelligent Reflective Surfaces (IRS) are reconfigurable metasurfaces designed to enhance wireless communication by passively reflecting electromagnetic waves towards desired directions, as illustrated in Fig. 7. Composed of numerous low-cost reflective elements, IRS can

dynamically control the phase of incident signals without requiring active transmission or significant power consumption [49].

IRS significantly improves signal propagation in challenging scenarios such as non-line-of-sight (NLOS) and deep-fade environments by increasing the signal to interference and noise ratio (SINR). These surfaces can be strategically deployed on buildings, vehicles, and other structures to enhance coverage and reliability without modifying existing infrastructure. Applications include beam steering, beamforming, and multi-user communication, with performance further enhanced through integration with machine learning techniques such as deep learning and neural networks.

By complementing massive MIMO, a key 5G technology, IRS contributes to improved spectral efficiency and higher data rates, positioning it as an important component in future 6G networks [46].

4.11. Wireless energy transmission

Wireless Energy Transmission (WET) in 6G enables sustainable and wireless power delivery to devices like smartphones and IoT sensors. It reduces battery waste, extends device durability, simplifies maintenance, and lowers emissions. Integrated with Wireless Information and Energy Transmission (WIET), 6G base stations can transfer both data and electricity, supporting battery-free devices and enhancing IoT network efficiency and sustainability [50], as illustrated in Fig. 8.

4.12. Edge intelligence

Edge intelligence integrates AI with edge computing to enable real-time decision-making near data sources, reducing latency and bandwidth demands. This approach supports critical applications like autonomous vehicles, drones, augmented reality, and telesurgery, where milliseconds matter. Powered by 6G's ultra-low latency, high data rates, and scalability, edge intelligence enhances data privacy, computational efficiency, and seamless connectivity, marking a transformative advancement for intelligent systems. Fig. 9 presents the flowchart of edge computing [51].